

TECHNOLOGY STACK

Strategic Product Placement Analysis

Overview

The technology stack combines data visualization, web development, styling, and development tools to build an interactive retail analytics solution. Each technology has a clear role in analyzing product placement data and presenting Tableau insights through a Flask-based web application.

TABLEAU PUBLIC	Data Visualization and Analytics Used to build interactive charts, dashboard filters, calculated comparisons, and the analytical story. Published Tableau views are embedded in the web application.
PYTHON	Application Runtime Provides the programming environment used to run the Flask web application and serve the project interface.
FLASK	Web Application Framework Used to create the web application, define the home route, and render the HTML template containing the project interface and Tableau embeds.
HTML	Web Page Structure Defines the structure of the Home, About, Dashboard, Story, and Project Information sections.
CSS	Interface Styling Controls the visual presentation, spacing, layout, backgrounds, typography, and responsive appearance of the web interface.
BOOTSTRAP	Responsive Frontend Framework Supports responsive layout, navigation, containers, grid structure, and reusable interface components.
JAVASCRIPT	Frontend Interaction Supports template behavior, visual effects, navigation interactions, and embedded visualization loading where required.
VS CODE	Development Environment Used to organize project folders, edit Flask and HTML files, manage static assets, and run the project through the integrated terminal.

TECHNOLOGY INTEGRATION

How the Stack Works Together

Layer	Technology	Project Role
Analytics Layer	Tableau Public	Analyzes retail data and creates the dashboard and story.
Backend Layer	Python + Flask	Runs the web application and renders the project template.
Structure Layer	HTML	Organizes the website sections and Tableau embed areas.
Presentation Layer	CSS + Bootstrap	Creates a clean, responsive, and consistent user interface.
Interaction Layer	JavaScript	Supports frontend interactions and visualization loading.
Development Layer	VS Code	Provides the environment for editing, organizing, and running the project.

Technology Flow

Retail Dataset → Tableau Public → Dashboard and Story → Flask Application → Bootstrap Web Interface → User Insights

Tableau performs the visual analysis and publishes the dashboard and story. Flask serves the web application, while HTML, CSS, Bootstrap, and JavaScript create the project interface and support the embedded visual experience.

Technology Stack Outcome

The selected technology stack is suitable for the project because it separates data visualization from web presentation while keeping the integration simple. Tableau Public provides interactive retail analytics, Flask serves the application, and Bootstrap-based frontend technologies present the dashboard and story in a clean and accessible format.

Team Lead

bhavyagogeneni@gmail.com

Members

chilakanagapravallika@gmail.com

karramaniroshan@gmail.com

naaperusandeep333@gmail.com

kasturibhanu162@gmail.com